



Genesis Biotech Inc.

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Fibronectin

Catalog No. PG-40004

Quantity: 500 µg

Protein species: cow or pig

Cytological test: cell spread rate should be over 80%

Applications tested: cell culture

Type: purified protein from blood

Form: lyophilized protein

Target description

Fibronectin (FN) is involved in many cellular processes, including tissue repair, embryogenesis, blood clotting, and cell migration/adhesion. Fibronectin exists in two main forms: 1) insoluble glycoprotein dimer that serves as a linker in the extracellular matrix, 2) soluble disulphide linked dimer found in the plasma. The plasma form is synthesized by hepatocytes, and the ECM form is made by fibroblasts, chondrocytes, endothelial cells, macrophages, as well as certain epithelial cells. Fibronectin sometimes serves as a general cell adhesion molecule by anchoring cells to collagen or proteoglycan substrates. Fibronectin also can serve to organize cellular interaction with the ECM by binding to different components of the extracellular matrix and to membrane-bound Fibronectin receptors on cell surfaces.

Potential applications

1. Fibronectin plays an important role in wound healing: Studies carried out about wound healing after topical application of FN in skin wound in rats, in the corneal epithelial wound in rabbits and periodontal surgical wound in patients suggest a beneficial FN action in the healing process.
2. Cosmetics: Fibronectin is involved in many cellular processes including tissue repair, and serves as a general cell adhesion molecule by anchoring cells to collagen. Recently, it was used for cosmetics.

Comments

Fibronectin with molecular mass 230kDa is purified from adult cow or pig blood. By electrophoresis method. Use non-reducing SDS-PAGE method. Sample adding quantity should not be less than 10 µg, stain with Coomassie Blue R-250. Purity should not be lower than 90.0% by scanning.

Related Products

1. IGF-1 (insulin-like growth factor-1) (PG-40001)
2. PDGF-B (platelet-derived growth factor B-chain) (PG-40002).
3. bFGF (basic fibroblast growth factor) (PG-40003).
4. TNF α (Tumor necrosis factor-alpha) (PG-40005).

Antibiotic residue

No activity of ampicillin or other antibiotic residue detected.

Bacterial endotoxin content

Bacterial endotoxin content is not higher than 0.1ng/ug (1EU/ug)

Biological Activity

Cytological Test: tested with Vero cells, cell spread rate should be over 80%

Solubility

It is recommended to reconstitute the lyophilized Fibronectin in sterile H₂O not less than 100µg/ml, which can then be further diluted to other aqueous solutions.

Storage

For short-term use store at 4 °C. For long-term storage keep at -20°C.

References

1. Potts JR and Campbell ID. .Structure and Function of Fibronectin Modules. *Matrix Bio.* 15: 313-320, 1996.
2. Chen ZY, Maricq HR, Pilia PA, Dobson RL, Silver RM, Ainsworth SK. Fibronectin distribution in nailfold biopsies of scleroderma (systemic sclerosis) patients. *Acta Derm Venereol.* 65(3): 185-9, 1985.

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